CYBER SECURITY INTERNSHIP REPORT

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GitHub Repositories:

- FUTURE_CS_01
- FUTURE_CS_02
- FUTURE_CS_03

Submission Date: (Fill Here)

TASK 1 — WEB APPLICATION SECURITY TESTING

1.1 Objective

The objective of this task was to perform vulnerability assessment on a web application using OWASP Top 10 methodology. The goal was to identify common web vulnerabilities such as SQL Injection, Cross Site Scripting (XSS), authentication weaknesses, and security misconfigurations using professional penetration testing tools.

1.2 Tools Used

The following tools were used during the security assessment:

- Kali Linux

- Burp Suite
- OWASP ZAP
- Nmap
- PortSwigger Web Security Academy
- DVWA (Damn Vulnerable Web Application)

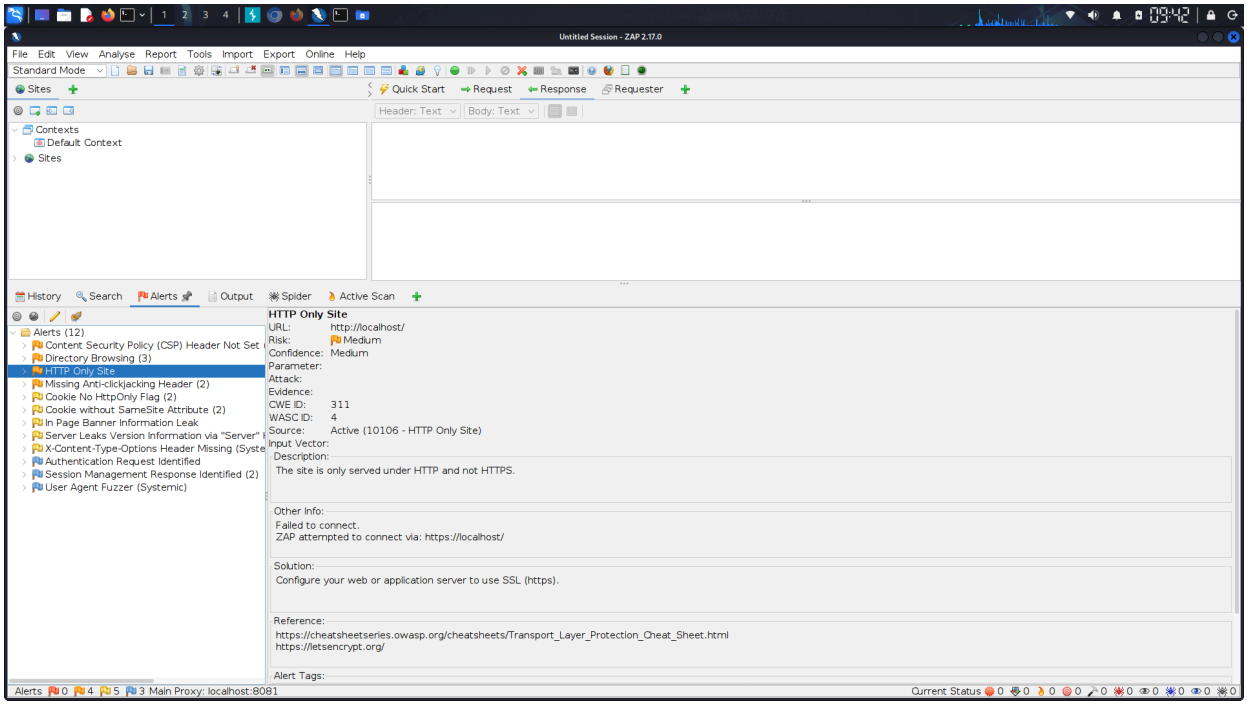
1.3 Environment Setup

A penetration testing environment was prepared using Kali Linux. DVWA was deployed locally using Docker to simulate a vulnerable web application. Burp Suite was configured as a proxy to intercept HTTP requests from the browser.

This environment allowed safe testing of vulnerabilities without affecting real systems.

📷 **Screenshot — ZAP Medium Risk Detection**

IMAGE PLACEHOLDER:



1.4 SQL Injection Testing

SQL Injection is one of the most critical vulnerabilities in web applications. It occurs when user input is directly inserted into database queries without proper validation or sanitization.

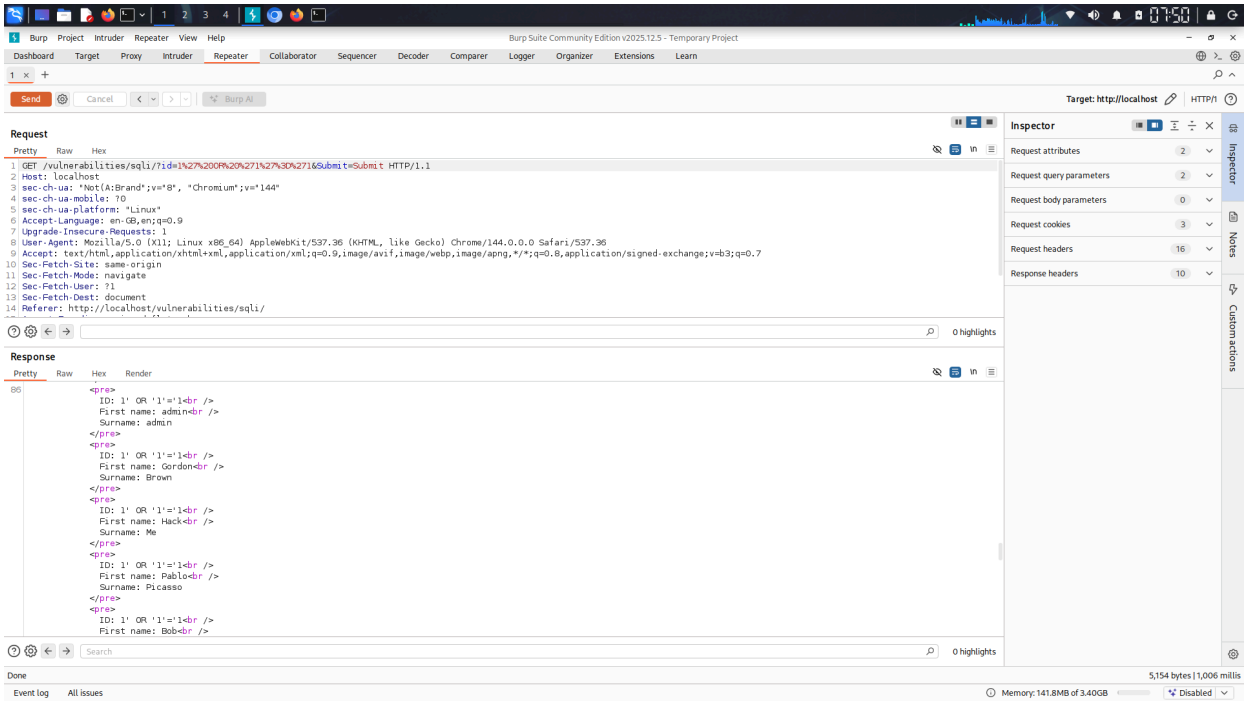
During testing, SQL Injection payloads were injected into input fields to manipulate backend database queries.

Example payload used:

' OR '1'='1

This payload forces the SQL query to always evaluate as true, allowing unauthorized access or retrieval of database records.

📷 Screenshot — SQL Injection Proof



The results showed multiple database records being returned, confirming the presence of a SQL Injection vulnerability.

1.5 Cross Site Scripting (XSS) Testing

Cross Site Scripting allows attackers to inject malicious JavaScript code into web pages that execute in a victim's browser.

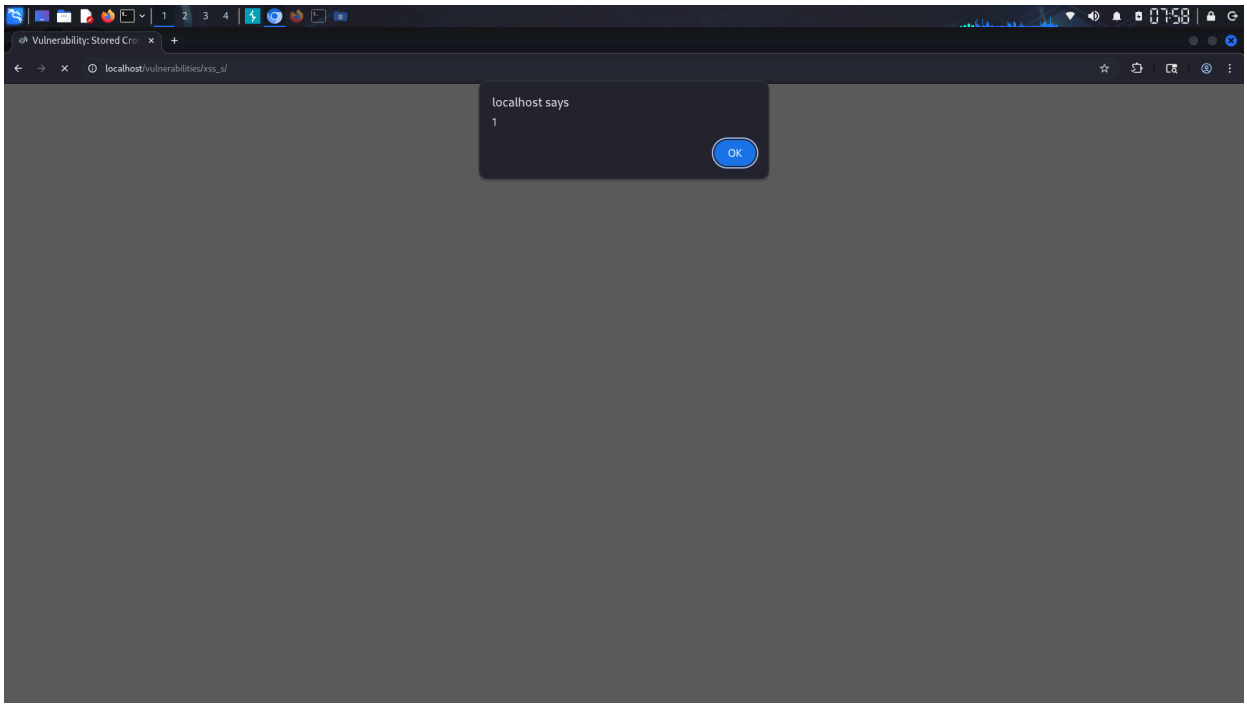
Testing was performed using both **Stored XSS** and **Reflected XSS** attacks.

Example payload used:

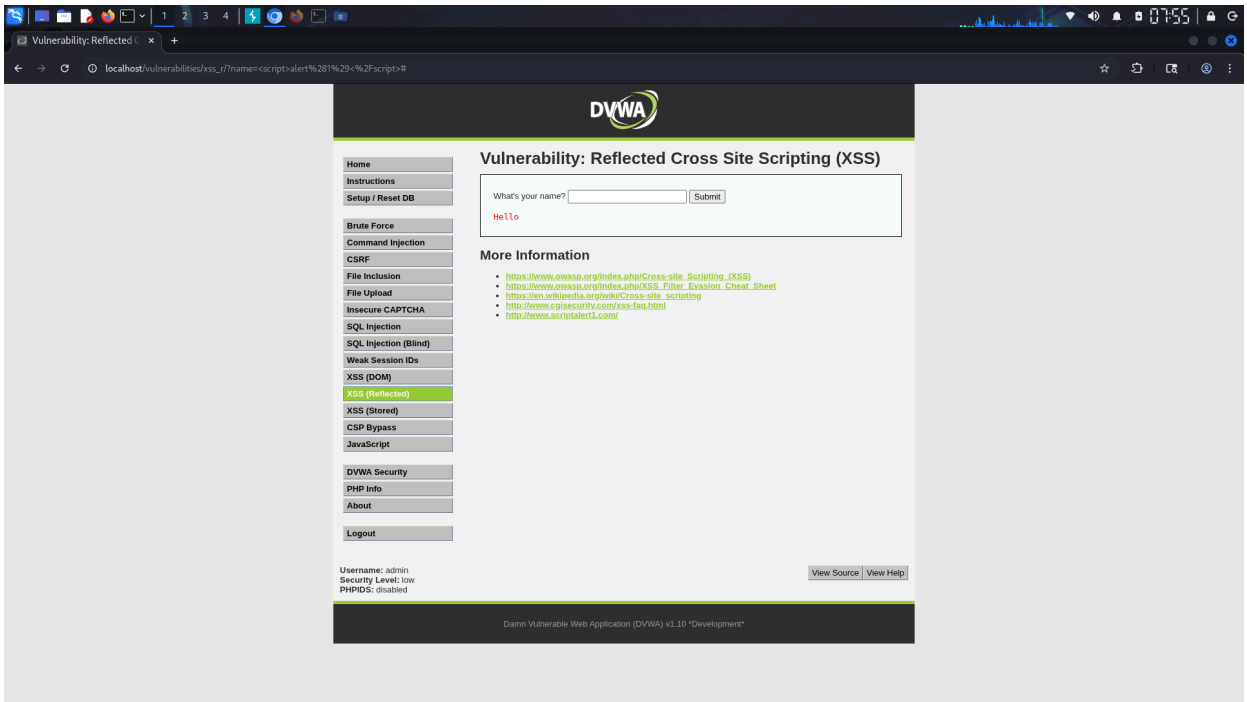
<script>alert(1)</script>

If the script executes in the browser, it confirms that the application does not properly sanitize user input.

📷 Stored XSS Proof



📷 Reflected XSS Proof



The successful execution of JavaScript alerts confirmed the presence of Cross Site Scripting vulnerabilities.

1.6 Authentication Weakness Testing

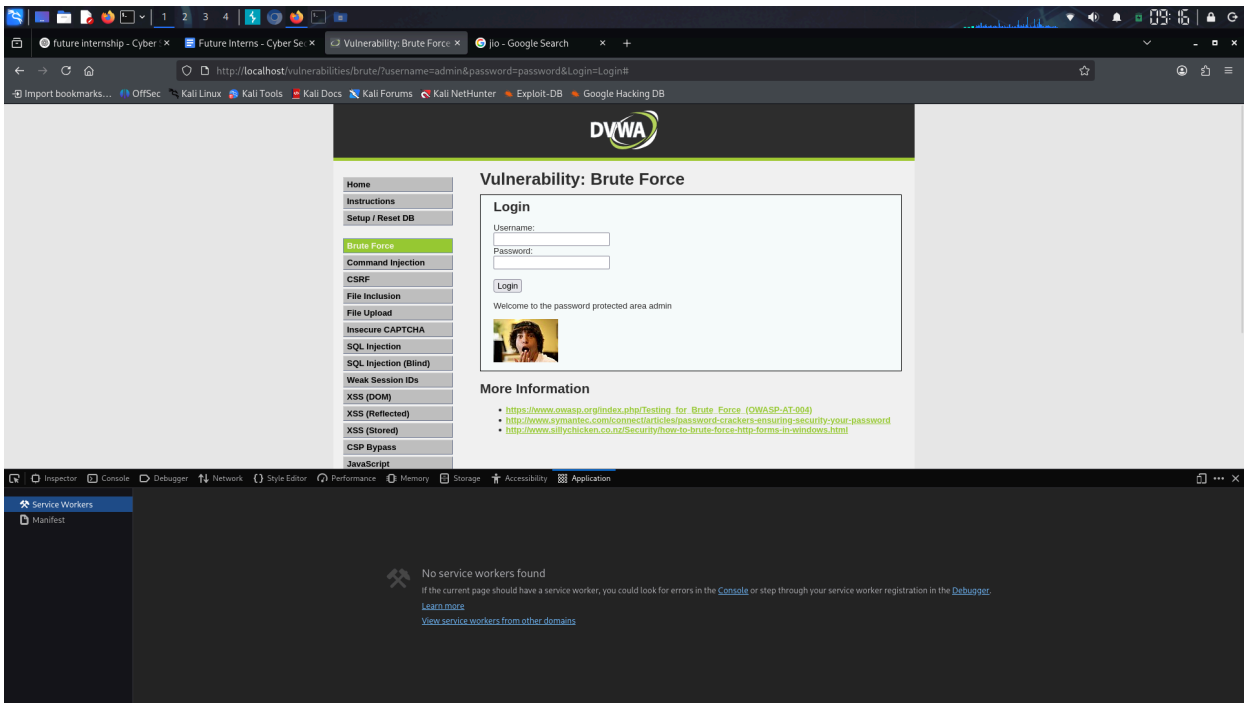
Authentication mechanisms were tested for common weaknesses such as weak credentials and lack of login protection.

Testing revealed that default credentials were accepted by the application.

Example credentials:

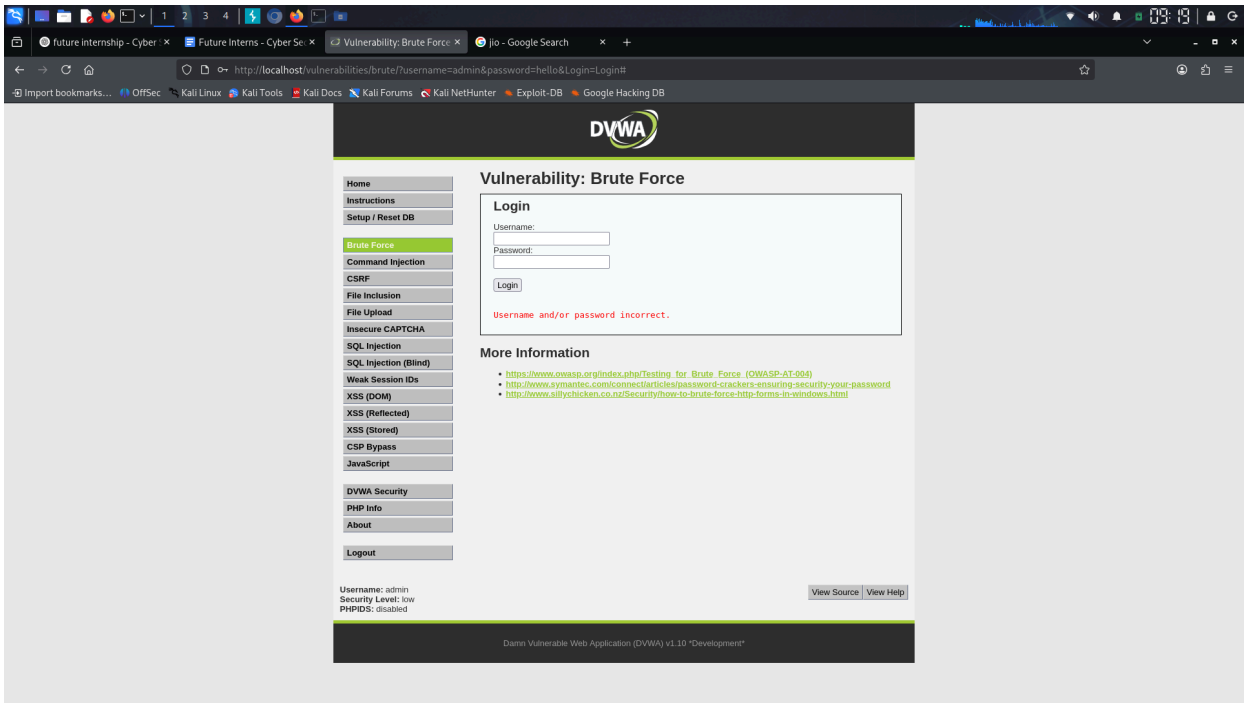
Username: admin
Password: password

📷 Default Credentials Accepted



Additionally, multiple login attempts were allowed without account lockout protection.

📷 No Login Lockout Mechanism



This indicates a **Brute Force vulnerability**.

1.7 Network Reconnaissance using Nmap

Network reconnaissance was performed using Nmap to identify open ports and running services.

Example command used:

nmap -sV scanme.nmap.org

This command performs service version detection.

📷 Nmap Scan Result

```
┌──(iamin㉿iamin)-[~]
└─$ nslookup uptoskills.com
Server:         10.246.104.94
Address:        10.246.104.94#53

Non-authoritative answer:
Name:   uptoskills.com
Address: 104.21.77.200
Name:   uptoskills.com
Address: 172.67.211.113
Name:   uptoskills.com
Address: 2606:4700:3035::ac43:d371
Name:   uptoskills.com
Address: 2606:4700:3033::6815:4dc8


┌──(iamin㉿iamin)-[~]
└─$ nmap 104.21.77.200
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 23:02 +0530
Stats: 0:00:01 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.55% done
Nmap scan report for 104.21.77.200
Host is up (0.19s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE SERVICE
80/tcp   open  http
443/tcp  open  https
8080/tcp open  http-proxy
8443/tcp open  https-alt
Nmap done: 1 IP address (1 host up) scanned in 16.90 seconds

┌──(iamin㉿iamin)-[~]
└─$ nmap 172.67.211.113
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 23:03 +0530
Nmap scan report for 172.67.211.113
Host is up (0.16s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE SERVICE
80/tcp   open  http
443/tcp  open  https
8080/tcp open  http-proxy
8443/tcp open  https-alt

Nmap done: 1 IP address (1 host up) scanned in 13.15 seconds
```

Additional scans were also conducted.

📷 Additional scan outputs:

```
┌──(iamin㉿iamin)-[~]
└─$ nmap -sV -p 80,443,8080,8443 uptoskills.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 23:09 +0530
Nmap scan report for uptoskills.com (104.21.77.200)
Host is up (0.10s latency).
Other addresses for uptoskills.com (not scanned): 172.67.211.113 2606:4700:3033::6815:4dc8 2606:4700:3035::ac43:d371

PORT     STATE SERVICE    VERSION
80/tcp   open  http       Cloudflare http proxy
443/tcp  open  ssl/http   Cloudflare http proxy
8080/tcp open  http-proxy cloudflare
8443/tcp open  ssl/http   Cloudflare http proxy
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at
https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port8080-TCP:V=7.98%I=7%D=2/26%Time=69A08575%P=x86_64-pc-linux-gnu%r(Ge
SF:tRequest,1AC,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nDate:\x20Thu,\x2026
SF:\x20Feb\x202026\x2017:40:05\x20GMT\r\nContent-Length:\x2068\r\nConnecti
SF:on:\x20close\r\nCache-Control:\x20private,\x20max-age=0,\x20no-store,\x
SF:20no-cache,\x20must-revalidate,\x20post-check=0,\x20pre-check=0\r\nRefe
SF:rrer-Policy:\x20same-origin\r\nExpires:\x20Thu,\x2001\x20Jan\x201970\x2
SF:000:00:01\x20GMT\r\nProxy-Status:\x20Cloudflare-Proxy;error=http_reques
SF:t_error\r\nCF-RAY:\x209d4139bf09bdfe8b-SIN\r\n\r\nCloudflare\x20encount
SF:ered\x20an\x20error\x20processing\x20this\x20request:\x20Bad\x20Request
SF:")%r(HTTPOptions,1AC,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nDate:\x20Th
SF:u,\x2026\x20Feb\x202026\x2017:40:06\x20GMT\r\nContent-Length:\x2068\r\n
SF:Connection:\x20close\r\nCache-Control:\x20private,\x20max-age=0,\x20no-
SF:store,\x20no-cache,\x20must-revalidate,\x20post-check=0,\x20pre-check=0
SF:\r\nReferrer-Policy:\x20same-origin\r\nExpires:\x20Thu,\x2001\x20Jan\x2
SF:01970\x2000:00:01\x20GMT\r\nProxy-Status:\x20Cloudflare-Proxy;error=htt
SF:p_request_error\r\nCF-RAY:\x209d4139c19ea3069f-HKG\r\n\r\nCloudflare\x2
SF:0encountered\x20an\x20error\x20processing\x20this\x20request:\x20Bad\x2
SF:0Request")%r(RTSPRequest,9B,"<html>\r\n<head><title>400\x20Bad\x20Reque
SF:st</title></head>\r\n<body>\r\n<center><h1>400\x20Bad\x20Request</h1></
SF:center>\r\n<hr><center>cloudflare</center>\r\n</body>\r\n</html>\r\n")%
SF:r(FourOhFourRequest,1AC,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nDate:\x2
SF:0Thu,\x2026\x20Feb\x202026\x2017:40:06\x20GMT\r\nContent-Length:\x2068\
SF:r\nConnection:\x20close\r\nCache-Control:\x20private,\x20max-age=0,\x20
SF:no-store,\x20no-cache,\x20must-revalidate,\x20post-check=0,\x20pre-chec
SF:k=0\r\nReferrer-Policy:\x20same-origin\r\nExpires:\x20Thu,\x2001\x20Jan
SF:\x201970\x2000:00:01\x20GMT\r\nProxy-Status:\x20Cloudflare-Proxy;error=
SF:http_request_error\r\nCF-RAY:\x209d4139c55db3c5e8-HKG\r\n\r\nCloudflare
SF:\x20encountered\x20an\x20error\x20processing\x20this\x20request:\x20Bad
SF:\x20Request")%r(Socks5,13C,"HTTP/1\.1\x20400\x20Bad\x20Request\r\nServe
SF:r:\x20cloudflare\r\nDate:\x20Thu,\x2026\x20Feb\x202026\x2017:40:06\x20G
SF:MT\r\nContent-Type:\x20text/html\r\nContent-Length:\x20155\r\nConnectio
SF:n:\x20close\r\nCF-RAY:\x20-\r\n\r\n<html>\r\n<head><title>400\x20Bad\x2
SF:0Request</title></head>\r\n<body>\r\n<center><h1>400\x20Bad\x20Request<
SF:/h1></center>\r\n<hr><center>cloudflare</center>\r\n</body>\r\n</html>\
SF:r\n");

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.73 seconds
```

```
┌──(iamin㉿iamin)-[~]
└─$ nmap -p- -T4 uptoskills.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 23:20 +0530
Nmap scan report for uptoskills.com (104.21.77.200)
Host is up (0.17s latency).
Other addresses for uptoskills.com (not scanned): 172.67.211.113 2606:4700:3033::6815:4dc8 2606:4700:3035::ac43:d371
Not shown: 65473 filtered tcp ports (no-response), 49 filtered tcp ports (admin-prohibited)
PORT     STATE SERVICE
80/tcp   open  http
443/tcp  open  https
2052/tcp open  clearvisn
2053/tcp open  knetd
2082/tcp open  infowave
2083/tcp open  radsec
2086/tcp open  gnunet
2087/tcp open  eli
2095/tcp open  nbx-ser
2096/tcp open  nbx-dir
8080/tcp open  http-proxy
8443/tcp open  https-alt
8880/tcp open  cddbp-alt

Nmap done: 1 IP address (1 host up) scanned in 411.14 seconds
```

These scans helped identify exposed services and potential attack surfaces.

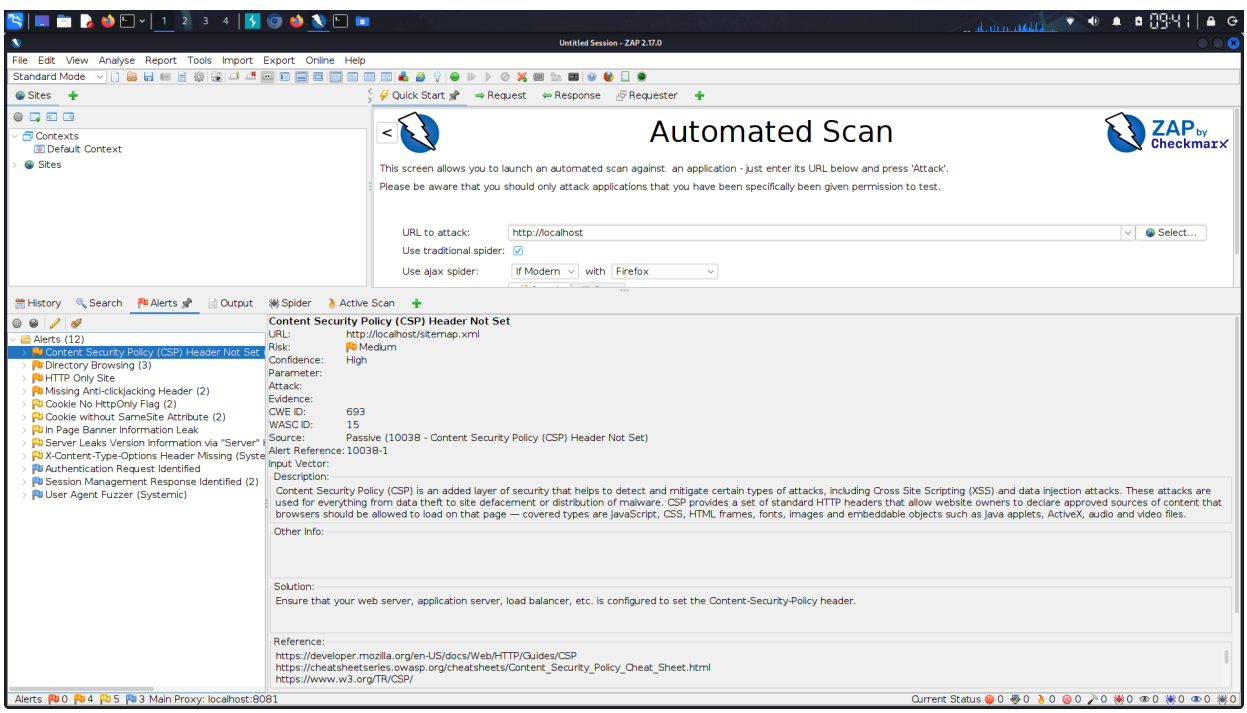
1.8 OWASP ZAP Security Scan

Automated vulnerability scanning was performed using OWASP ZAP.

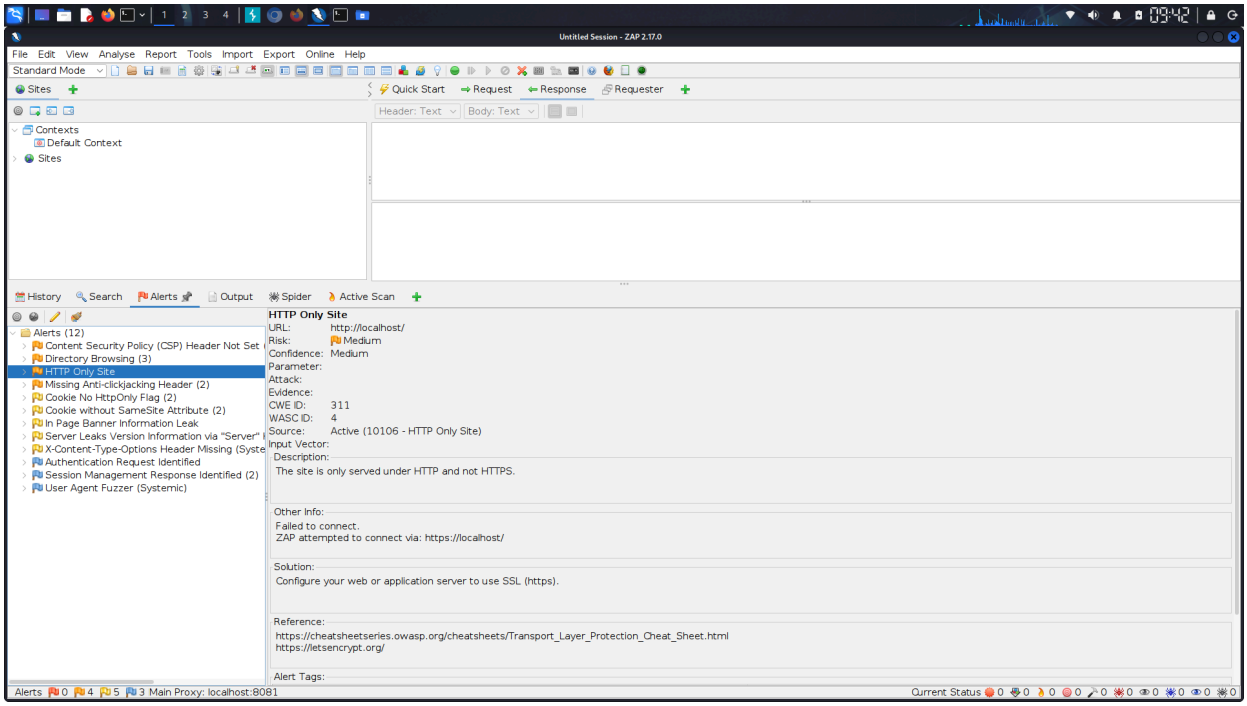
ZAP identified multiple security issues including:

- Missing Security Headers
- Directory Listing Enabled
- Cookie Security Flags Missing

📷 ZAP Alerts Overview



📷 ZAP Medium Risk Detection



1.9 OWASP Top 10 Vulnerability Mapping

Vulnerability	OWASP Category	Severity	Mitigation
SQL Injection	A03:2021 Injection	Critical	Use parameterized queries
Cross Site Scripting	A03:2021 Injection	High	Input validation and output encoding
Weak Authentication	A07:2021 Auth Failures	High	Implement strong password policy
Security Misconfiguration	A05:2021	Medium	Disable unnecessary features

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1.10 Task 1 Summary

Total vulnerabilities identified:

Severity	Count
Critical	1
High	2
Medium	1

Overall risk level of the tested application was **High**.

GitHub Repository:

FUTURE_CS_01